



Effects of Al and Si addition on the structure and properties of CoFeNi equal atomic ratio alloy



T.T. Zuo^a, R.B. Li^b, X.J. Ren^c, Y. Zhang^{a,*}

^a State Key Laboratory for Advanced Metals and Materials, University of Science and Technology Beijing, Beijing 100083, China

^b School of Mechanical Engineering, Shanghai Dianji University, Shanghai 200245, China

^c School of Engineering, Liverpool John Moores University, Liverpool 611310, United Kingdom

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ABSTRACT

In this work, a series of Al_xCoFeNi and CoFeNiSi_x high-entropy alloys (HEAs) of different Al and Si molar ratio ($x=0, 0.25, 0.5, 0.75$ and 1) were designed and, the effects of Al and Si addition on the structure and properties of the materials was investigated by a systematic study on the phase, microstructure, mechanical behavior, electrical and magnetic properties. It was found that the amounts of Al and Si additions strongly influence the phase structures of the alloys; high molar ratio of Al element can change the FCC structure to BCC structure, while more Si addition can lead to new compounds. Both Al and Si addition can increase the yield strength and hardness with the sacrifice of plasticity and, the effect of adding Si on the mechanical properties is more significant than that of adding Al. Characterization of the magnetic properties and hysteresis loops revealed that, all these alloys show typical ferromagnetic behavior. The saturation magnetization decreases from 151.3 emu/g ($x=0$) to 101.8 emu/g ($x=1$) when changing the Al content; and decreases from 151.3 emu/g ($x=0$) to 80.5 emu/g ($x=0.75$) with changing the Si content. Si addition can decrease the saturation magnetization more significantly than Al addition. The opposite trend can be found in the effects of the alloying element on the electrical resistivity with varying Al or Si content, adding Si can increase the electrical resistivity from $16.7 \mu\Omega \text{ cm}$ to $82.89 \mu\Omega \text{ cm}$. It was also found that, the alloys also undergo very small magnetostriction, which is essential to ensure that the materials are not stressed when an external magnetic field is applied (or conversely, that external stresses do not disrupt the magnetic properties).

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1. Introduction

The high entropy alloy (HEA), sometimes called multi-component alloy, was originally discovered by Cantor and Yeh [1,2]. It was defined as alloys that contain more than five principal elements in equal or near equal atomic percent (at%), and the atomic ratio of each element is between 5 and 35% [2]. Recent work has suggested that this material group can be extended to include the quaternary alloy system [3]. The observation of unique FCC and/or BCC solid solutions structures together with other interesting properties such as high strength, high hardness, good resistances to wear, high thermal stability, high electrical resistivity, and good hydrogen storage properties [4–13], has attracted strong research interest on these particular alloy systems. As a result, a large amount of knowledge has been acquired on the fundamentals of the materials and their applications. However,

almost all of the researches have been mainly focused on the mechanical performances, while systematic work on the understanding of the physical properties [8,14,15] is still limited. An integrated study of the effects of alloying element on the mechanical and physical properties is essential to establish the framework for further developments and applications of the material system. HEAs possess high electrical resistivity, which is required for applications as soft magnetic materials. In our previous study, it was found that FeCoNi(AlSi)_x ($0 \leq x \leq 0.8$ in molar ratio) HEAs exhibit excellent magnetic properties [16]. Accompanying the compositional and structural changes, when increasing the Al and Si content simultaneously, the saturation magnetization decreases from 1.31 T (at $x=0$) to 0.46 T (at $x=0.8$), while the electrical resistivity reaches its peak value ($79.7 \mu\Omega \text{ cm}$) at $x=0.3$. However, limited work has been done to explore the role of each individual element (Al or Si) on this system, which is important for further materials development and composition design. In this paper, we will investigate the influence of Al and Si on the structure and properties of Al_xCoFeNi and CoFeNiSi_x high entropy alloy system, respectively.

* Corresponding author. Tel.: +86 10 62333073; fax: +86 10 62333447.

E-mail addresses: zuott1986.520@163.com (T.T. Zuo), lirb@sdju.edu.cn (R.B. Li), X.J.Ren@ljmu.ac.uk (X.J. Ren), drzhangy@ustb.edu.cn (Y. Zhang).

2. Experiment procedures

Alloy ingots with nominal compositions of Al_xCoFeNi ($x=0, 0.25, 0.5, 0.75$ and 1 , in molar ratios, denoted as Al0 , Al0.25 , Al0.5 , Al0.75 , and Al1 , respectively) and CoFeNiSi_x ($x=0, 0.25, 0.5, 0.75$, in molar ratios, denoted as Si0 , Si0.25 , Si0.5 , Si0.75 , respectively) were prepared by arc-melting of pure single element material with purity higher than 99.9 wt% (wt%) under a Ti-gettered high-purity argon atmosphere on a water-cooled Cu hearth. The alloys were remelted four times and flipped each time in order to improve the homogeneity of the material. The prepared alloy buttons were cut into appropriate forms for investigating their microstructure and compressive properties. Crystal structures of the materials were identified using an X-ray diffractometer under radiation conditions of 30 kV and 20 mA, with a Cu target and a scanning speed of $10^\circ/\text{min}$. Compressive tests of cylindrical samples with a dimension of $\phi 3 \times 6$ mm were performed using an MTS 809 materials testing machine at room temperature with a strain rate of $2 \times 10^{-4} \text{ s}^{-1}$. The hardness of the materials was measured using a Vickers hardness tester under a load of 200 g and a holding time of 15 s. The microstructures of the samples were examined by a ZEISS SUPRA 55 field emission scanning electron microscope (SEM) with the energy dispersive spectrometry (EDS). The magnetization curves were obtained on a vibrating sample magnetometer. The measurement of the magnetostrictive coefficient of the samples was conducted following an electrical resistance strain gauge method. The electrical resistivity was measured using a Precise Resistance Tester AT510.

3. Results and discussions

3.1. Crystal structure evolution

X-ray diffraction (XRD) measurements were employed to investigate the phase and structure of the as-cast samples. Fig. 1 (a) and (b) show the XRD patterns of the Al_xCoFeNi and CoFeNiSi_x alloy specimens, respectively. It is clearly shown that, without Al and Si addition ($x=0$), the alloy (Al0/Si0) possesses a single FCC structure. With increasing Al content, the atomic radius difference (δ) increases (Table 1), representing more significant lattice distortion. When x reaches 0.5, BCC phase appears and the intensity of the BCC peaks are enhanced with further increasing the Al content, as shown in Fig. 1, only BCC phase was detected when x reaches 1.0. The small diffraction peaks at low angle region ($2\theta \approx 30^\circ$) indicate the appearance of ordered BCC phase. The corresponding lattice constants of the alloys with different level of Al/Si addition have been calculated according to $d = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$ by identifying the strongest peaks in the XRD patterns, the results are listed in Table 2. It is clearly shown that the lattice constant of both FCC and BCC structures has increased with increasing Al content. As Al has much larger atom radius than other elements, the addition of Al cause the expansion of lattice and, the structure changes. As shown in Fig. 1(b), the XRD patterns of samples with Si addition exhibit a different variation trend. With increasing Si content, the matrix phase remains as a FCC structure and the lattice distortion caused by adding of Si is not as serious as the case for Al. This is probably associated with the fact that atomic size difference between Si and Co, Fe, Ni is not as significant as that for Al atom, so there is no need for the matrix to adjust the structure in order to release the distortion energy. But, there is a clear index peak for Ni_3Si when x is equal to 0.5 indicating that the phase has precipitated from the matrix, which has absorbed a significant amount of Co and Fe; the peak intensity of Ni_3Si phase for the sample with higher Si addition ($x=0.75$) becomes much stronger,

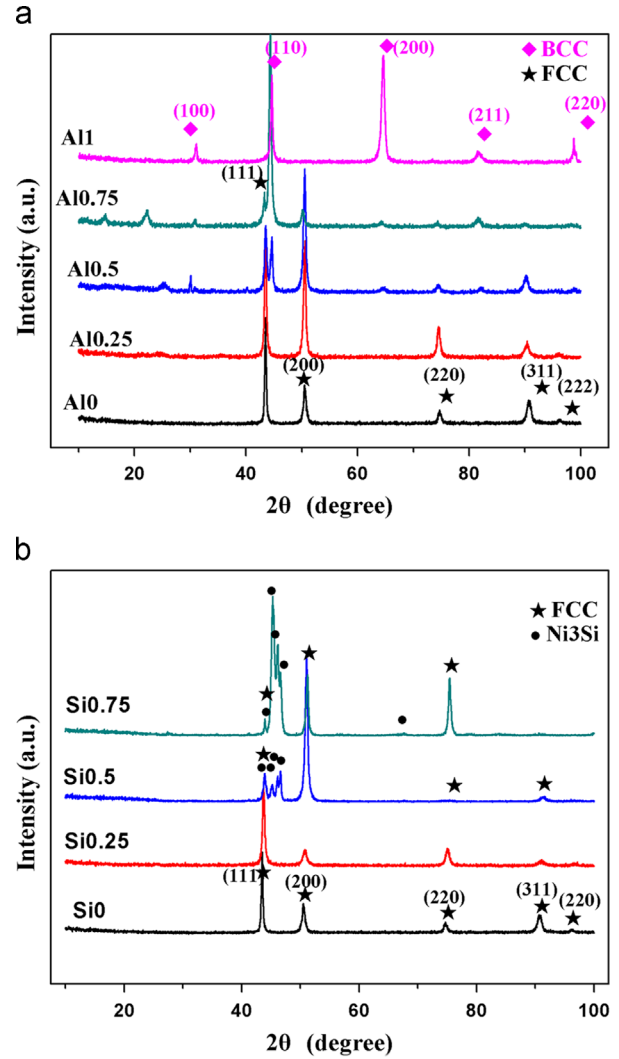


Fig. 1. XRD patterns of (a) Al_xCoFeNi and (b) CoFeNiSi_x alloys.

Table 1

The atomic radius difference (δ), Electronegativity difference ($\Delta\chi$), Valence electron concentration (VEC), Ω parameter and structure of the as-cast Al_xCoFeNi and CoFeNiSi_x ($x=0, 0.25, 0.5, 0.75$, and 1) alloy systems (atomic percent).

	δ	$\Delta\chi$	VEC	Ω	Structure
CoFeNi (Al0/Si0)	0.75	0.00330	9.00	12.12	FCC
Al _{0.25} CoFeNi (Al0.25)	3.706	0.06770	8.54	3.01	FCC
Al _{0.5} CoFeNi (Al0.5)	4.782	0.09700	8.14	1.96	FCC + BCC
Al _{0.75} CoFeNi (Al0.75)	5.599	0.10930	7.80	1.54	FCC + BCC
AlCoFeNi (Al1)	5.808	0.11755	7.50	1.31	BCC
CoFeNiSi _{0.25} (Si0.25)	1.980	0.03229	8.62	1.59	FCC
CoFeNiSi _{0.5} (Si0.5)	2.537	0.03194	9.29	1.03	FCC + Ni ₃ Si
CoFeNiSi _{0.75} (Si0.75)	3.221	0.03138	8.00	0.81	FCC + Ni ₃ Si

which indicates a larger amount of Ni_3Si precipitation. The increase of Ni_3Si phase with Si content may be attributed to the more negative mixing enthalpy between Ni and Si. Different from influence of Al, adding smaller atomic sized element Si leads to lattice contraction, the higher the content of silicon is, the smaller the lattice constant becomes (Table 2).

For conventional alloys, according to the Gibbs phase rule, $f=c-p+1$ (f : degree of freedom, c : number of elements, and p : number of existing phase), the maximum number of equilibrium phases in a 'c' component system is 'c+1'. When the material is solidified under non-equilibrium condition, the

number of phases could be larger than ‘ $c+1$ ’. Evident from the data in Fig. 1 and Tables 1 and 2, the number of phases in the quaternary Al_xCoFeNi and CoFeNiSi_x HEAs is well below the maximum number calculated by the Gibbs phase rule. It is known that, HEAs possess significantly higher entropy of mixing (ΔS_{mix}) in the liquid state, which could effectively increase the extent of

confusion in alloy systems and facilitate simple solid-solution phase formation. The formation of solid solutions and compounds may be affected by many other factors such as chemical bond, dimension of atom, electron concentration, difference of electro-negativity, and so on. In a previous study, Zhang et al. have summarized [17] that small atomic size differences (δ) and near-zero values of the absolute enthalpy of mixing (ΔH_{mix}) facilitate the formation of solid-solutions of multi-component alloys, under condition where the atomic size differences is less than 6.5%, the enthalpy of mixing is in the range of -15 to 5 kJ/mol and the entropy of mixing (ΔS_{mix}) is in the range of 12 – 17.5 J/K mol. In another study, Yang et al. [18] further simplified this rule by proposing the parameter Ω ($\Omega = \Delta S_{\text{mix}} T_m / |\Delta H_{\text{mix}}|$), the condition for the high-entropy (HE) stabilized solid-solution is $\Omega \geq 1.1$ and $\delta \leq 6.6\%$. The calculated parameters Ω and δ of the Al_xCoFeNi alloys (Table 1) agrees with this rule well; while the CoFeNiSi_x alloys have smaller Ω than the critical value. All these results can be put as further evidence to prove the validity of the rule of solid solution formation. Recently, Guo et al. [19] reported that valence electron concentration, ($\text{VEC} = \sum_{i=1}^n c_i (\text{VEC})_i$), has a significant influence on phase stability. The work suggested that, in HEAs,

Table 2

Lattice constants of the as-cast Al_xCoFeNi and CoFeNiSi_x ($x=0, 0.25, 0.5, 0.75$, and 1) alloy systems (atomic percent).

Alloys	FCC (Å)	BCC(Å)
Al0/Si0	3.5990	
Al0.25	3.6004	
Al0.5	3.6090	2.8672
Al0.75	3.6162	2.8819
Al1		2.8824
Si0.25	3.5815	
Si0.5	3.5812	
Si0.75	3.5880	

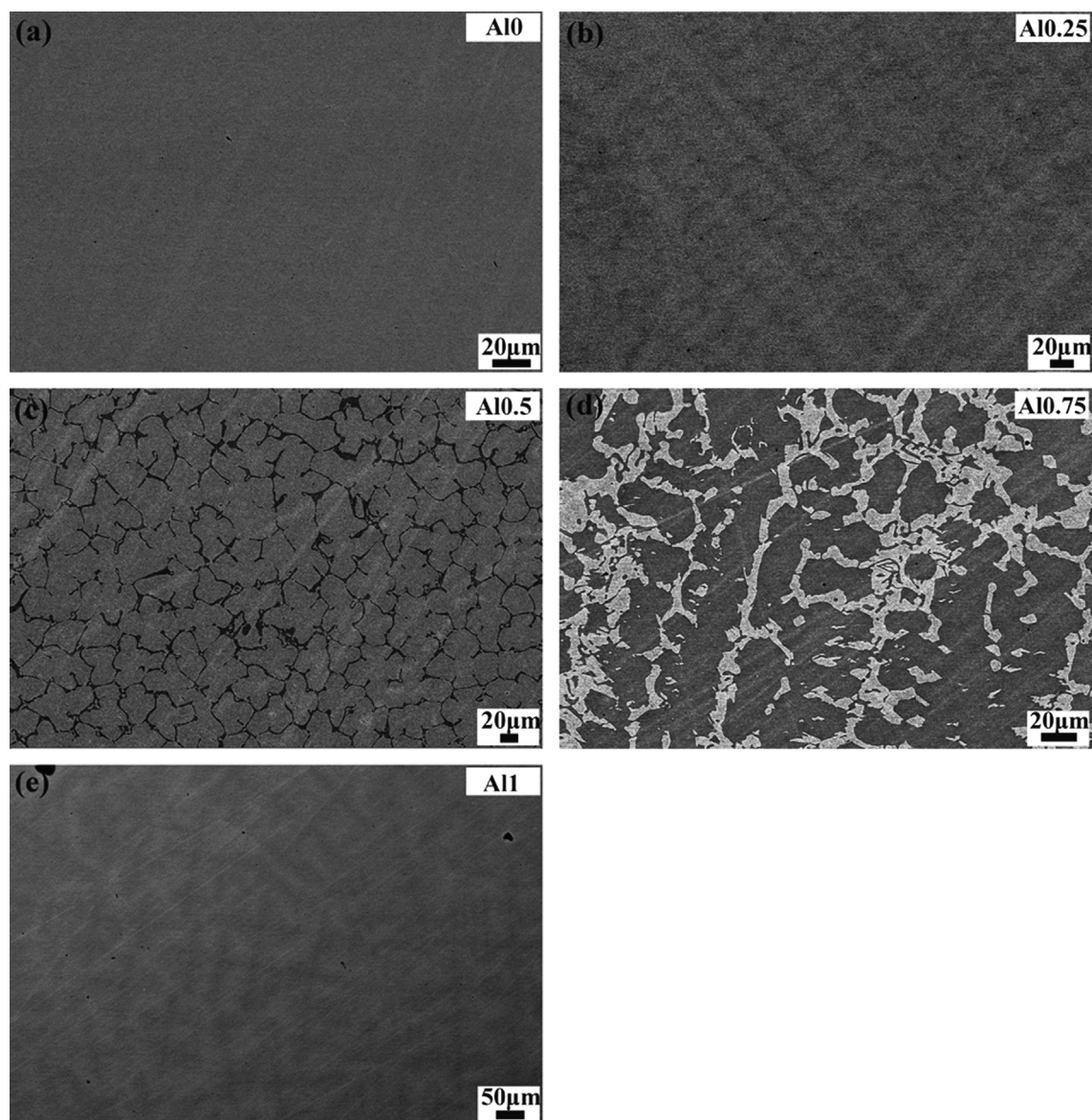


Fig. 2. SEM backscattering electron images of Al_xCoFeNi alloys, (a) $x=0$, (b) $x=0.25$, (c) $x=0.5$, (d) $x=0.75$, and (e) $x=1$.

single FCC phase exists at $VEC \geq 8.0$, mixed FCC and BCC phases will co-exist at $6.87 \leq VEC < 8.0$, and, sole BCC phase exists at $VEC < 6.87$. Evident from the VEC data in Table 1, our results agree with this rule very well. In conventional binary alloy, electro-negativity difference ($\Delta\chi = \sqrt{\sum_{i=1}^n c_i(\chi_i - \bar{\chi})^2}$) [19] plays an important role in solid solution formation. In the two alloy systems concerned in this work, the value of electro-negativity increases with increasing Al or Si content (Table 1). But no direct relationship can be found between the $Al_xCoFeNi$ and $Si_xCoFeNi$ alloy systems, which needs to be further studied.

3.2. Microstructure and chemical composition analysis

Fig. 2 shows the SEM backscattering electron image of $Al_xCoFeNi$ alloys. As shown in Fig. 2(a), the elements distribute uniformly in the specimen without Al addition, while element segregation is observed and the microstructure gradually turns to be typical dendrites with increasing Al content. As shown in Fig. 2(b)–(e), the morphology of the dendrites is different in alloys with different Al content. With lower Al content ($x=0.25$), the microstructure exhibits slender dendrites, then turns to flower-like dendrites at $x=0.5$, and herringbone dendrites at $x=0.75$. When x reaches 1, the dendrites structure is not so obvious any more, and elements tend to distribute uniformly, which is probably due to the relatively high mixing entropy of this quaternary alloy at equal atomic ratio. EDS analysis data (Table 3) shows that, the interdendritic area is rich in Al and Ni, this is reasonable as both elements have more negative mixing of enthalpy.

As shown in Fig. 3, the microstructure change of the $CoFeNiSi_x$ alloys is different and more complex when the Si content is high. At $x=0.5$, the microstructure was found to consist of three parts, and cubic Ni_3Si phase appears according to the XRD analysis (Fig. 1 (b)). The composition of the FCC matrix (marked as A in Fig. 3) is very close to the nominal composition of the alloy with slightly higher concentration of elements Fe and Co. TEM analysis has been performed to establish the detailed nature of the structure, some typical results are shown in Fig. 4. The TEM morphology in Fig. 4 (a) is consistent with the BSE results, the three different regions marked as 1, 2, 3 were corresponding to regions labeled as A, B, C in Fig. 3, respectively. Calibration of the diffraction spots (Fig. 4 (c) and (d)) revealed that, the interdendritic area is composed of two BCC phases, among which, region 2 is rich in Ni and Si, but region 3 is depleted of Ni. The results suggest that the interdendritic area is formed by spinodal decomposition. As these two phases have similar lattice constant, there is only one set of BCC diffraction peak visible on the XRD pattern. A similar phenomenon is observed in the $Si0.75$ alloy specimen. The block-like white part A is of FCC structure which is rich in Co and Fe. The interdendritic area rich in Ni and Si consists of two phases: the flocculent white phase (part B) and the black phase between them (part C). Based on the EDS analysis, it is clear that these two phases have similar Si content, but part B has more Ni element and part C has more Fe, Co elements. Compared with $Si0.5$ alloy, similar spinodal decomposition mechanism can be used to explain these observations.

3.3. Mechanical properties

Compressive engineering stress vs. engineering strain curves for the $Al_xCoFeNi$ and $CoFeNiSi_x$ alloys are plotted in Fig. 5. As it can be seen in Fig. 5(a), the stress strain curves of the $Al_xCoFeNi$ alloys varied with the Al content, which reflects that the phase structure plays an important role in compressive strength and plasticity in the material system, as BCC phase contributes more to the compressive yield strength than FCC phase. By increasing Al

Table 3

The EDS analysis of the as-cast $Al_xCoFeNi$ and $CoFeNiSi_x$ ($x=0, 0.25, 0.5, 0.75$, and 1) alloy systems (atomic percent).

Alloy	Region	Fe	Co	Ni	Al	Si
CoFeNi (Al0/Si0)	Nominal	33.333	33.333	33.333	/	/
	Total	35.510	31.810	32.680	/	/
Al _{0.25} CoFeNi (Al0.25)	Nominal	30.769	30.769	30.769	7.692	
	Total	30.45	30.73	29.19	9.63	
	DR	32.28	31.72	28.56	7.45	
	ID	29.77	29.09	30.17	10.98	
Al _{0.5} CoFeNi (Al0.5)	Nominal	28.571	28.571	28.571	14.286	/
	Total	27.81	27.49	27.28	17.42	
	DR	29.84	28.76	25.82	15.58	/
	ID	19.50	21.61	29.66	29.23	/
Al _{0.75} CoFeNi (Al0.75)	Nominal	26.667	26.667	26.667	20	
	Total	26.13	25.85	25.15	22.87	
	DR	31.77	28.60	24.29	15.34	
	ID	25.40	25.67	24.82	24.11	
AlCoFeNi (Al1)	Nominal	25	25	25	25	
	Total	24.38	23.92	23.28	28.42	
CoFeNiSi _{0.25} (Si0.25)	Nominal	30.769	30.769	30.769		7.692
	Total	30.89	30.52	30.27		8.31
	DR	32.43	32.25	28.32		7.00
	ID	26.24	26.58	32.98		14.20
CoFeNiSi _{0.5} (Si0.5)	Nominal	28.571	28.571	28.571		14.286
	Total	27.73	28.16	27.28		16.83
	A	30.04	29.03	26.52		14.41
	ID(B)	16.23	21.50	32.82		29.45
CoFeNiSi _{0.75} (Si0.75)	ID(C)	21.59	24.92	26.91		26.58
	Nominal	26.667	26.667	26.667		20
	Total	26.31	25.72	26.06		21.91
	A	31.95	28.57	23.33		16.15
	B	21.38	23.88	28.81		25.93
	C	32.08	31.07	11.65		25.20

content (x varies from 0.25 to 1), the yield strength increases from 158.4 MPa to 967.4 MPa with a reduction in the toughness when the phase structure varies from FCC to BCC. The best combination of strength and toughness can be achieved when x equals to 0.75, the fracture strength and the plastic deformation after fracture can reach 2787 MPa and 31.2%. Fig. 6 shows the morphology of the fractured surface of Al0.75 alloy, the small dimples and vein-like patterns suggests that the fracture mechanism of this alloy is pore/void coalescence and cleavage. As shown by the curve for Al1 alloy, no favorable effects are achieved by adding more Al element; although the yield strength has been increased, its fracture strength decreases significantly with no large scale of work hardening effect compared to the case of Al0.75 alloy and, the deformation after fracture can only reach 7.9%. A similar trend of mechanical property change is found in $CoFeNiSi_x$ alloys with varying Si content (Fig. 5(b)). The yield strength increases from 192.0 to 1297.1 MPa when x increased from 0.25 to 0.75. As shown in Fig. 7(a), the effect of Si addition on the strength increase is more significant than that of Al element. This is probably associated with the formation and distribution of the precipitation, which can significantly influence the mechanical properties of the alloy. Similar to silicon steel, higher Si content could lead to increased brittleness. As shown in Fig. 6(b), the fracture morphology of Si0.75 alloy exhibits typical fan-shaped cleavage feature, and micro-cracks can be clearly seen on the fractured surface. The hardness variation of the two series of alloys is similar to the yield strength, which rises with increasing Al and Si content. The maximum hardness is 450 HV_{0.2} for Al1 alloy and 568 HV_{0.2} for Si0.75 alloy.

3.4. Magnetic properties

Soft magnetic materials are, by definition, characterized by low coercivity. The most important characteristics desired for essentially

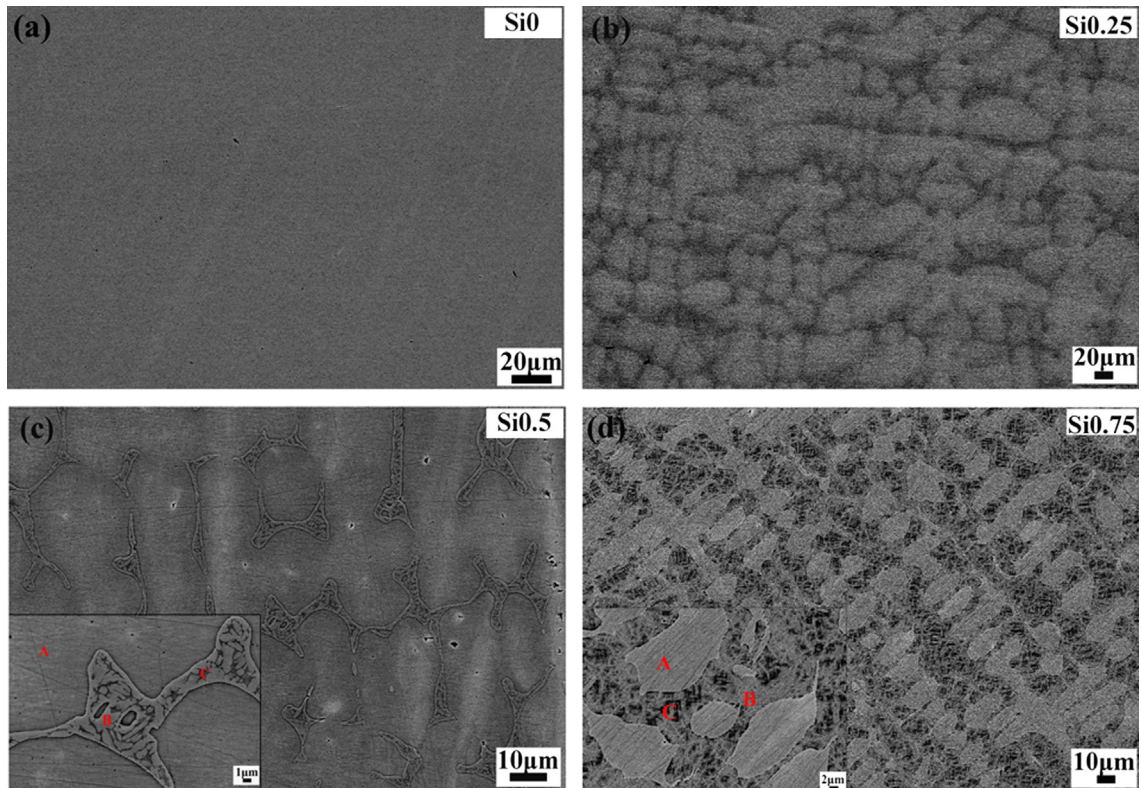


Fig. 3. SEM backscattering electron images of CoFeNiSi_x alloys, (a) $x=0$, (b) $x=0.25$, (c) $x=0.5$, (d) $x=0.75$.

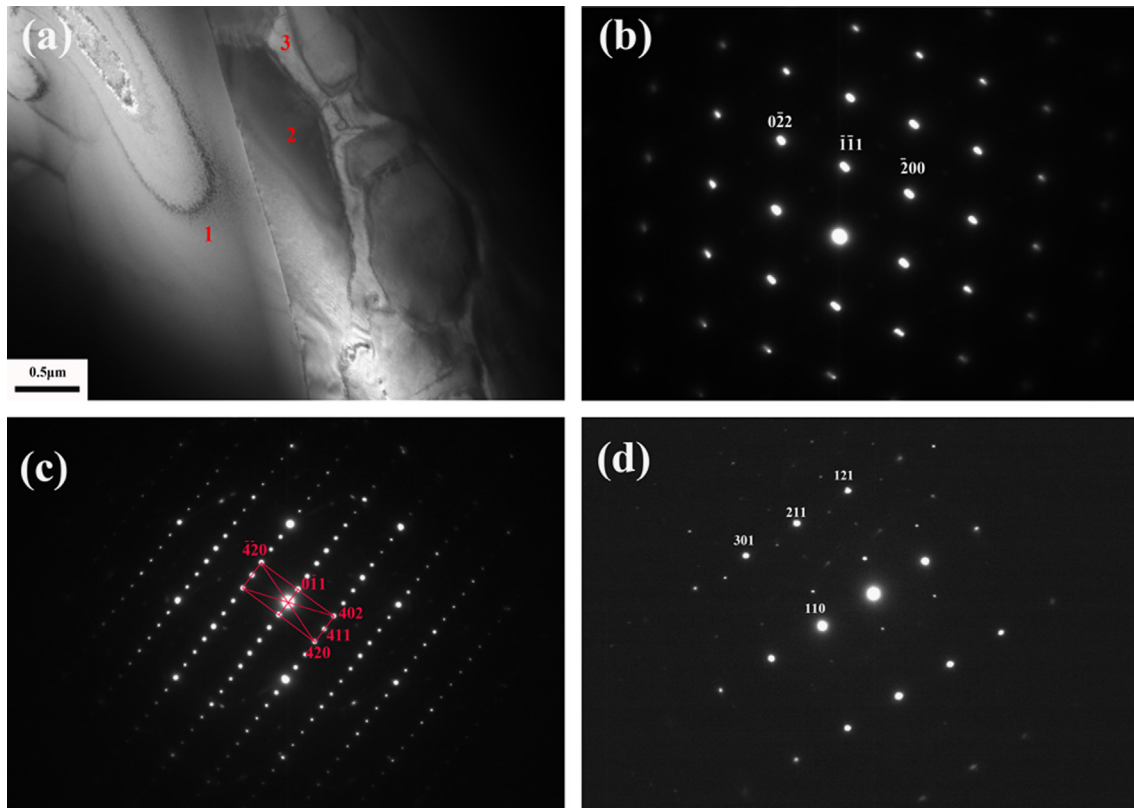


Fig. 4. TEM images of CoFeNiSi_{0.5} alloys (a) bright field image (b) electron diffraction pattern of region 1 from the $[0\ 1\ 1]$ zone axis, (c) electron diffraction pattern of region 2 from the $[1\ 2\ \bar{2}]$ zone axis, (d) electron diffraction pattern of region 3 from the $[1\ \bar{1}\ 3]$ zone axis.

all soft magnetic applications are high saturation induction, high permeability, low coercivity, and low core loss [20–22]. Fig. 8 shows the hysteresis loops of the Al_xCoFeNi and CoFeNiSi_x alloys. It is

evident that the saturation magnetization (M_s) decreases with increasing Al and Si content. Moreover, the decrease of M_s is more significant in the case of CoFeNiSi_x, for which the M_s value is reduced

from 149.65 emu/g to 80.57 emu/g when x increases from 0 to 0.75. Fig. 9 shows the variation of electrical resistivity of the alloys; different from the trend with M_s , the electrical resistivity (ρ) of the alloys increases with Al or Si addition. The maximum value of electrical resistance is obtained at $x=0.5$ for both alloy systems. The existence of the peak value is probably associated with the nature of the multiphase microstructure. Besides, the data also shows that the increase of electrical resistance with Si is more significant than that of Al, which is favorable as the increase of electrical

resistance of the material could effectively reduce the eddy loss. In addition, the new alloy systems also undergo very small magnetostriction, which is essential to ensure that the materials are not stressed when an external magnetic field is applied (or conversely, that external stresses do not disrupt the magnetic properties) [23,24]. This is another beneficial characteristic of the CoFeNiSi_x and Al_xCoFeNi alloys. As shown by the magnetostrictive curves of the two alloy systems presented in Fig. 10, the saturation magnetostriction coefficient (λ_s) of CoFeNiSi_x alloys is smaller than that of Al_xCoFeNi alloys, and in both alloy systems, λ_s are lower than 20 ppm. It is well established that, for soft magnetic materials, the intrinsic magnetic

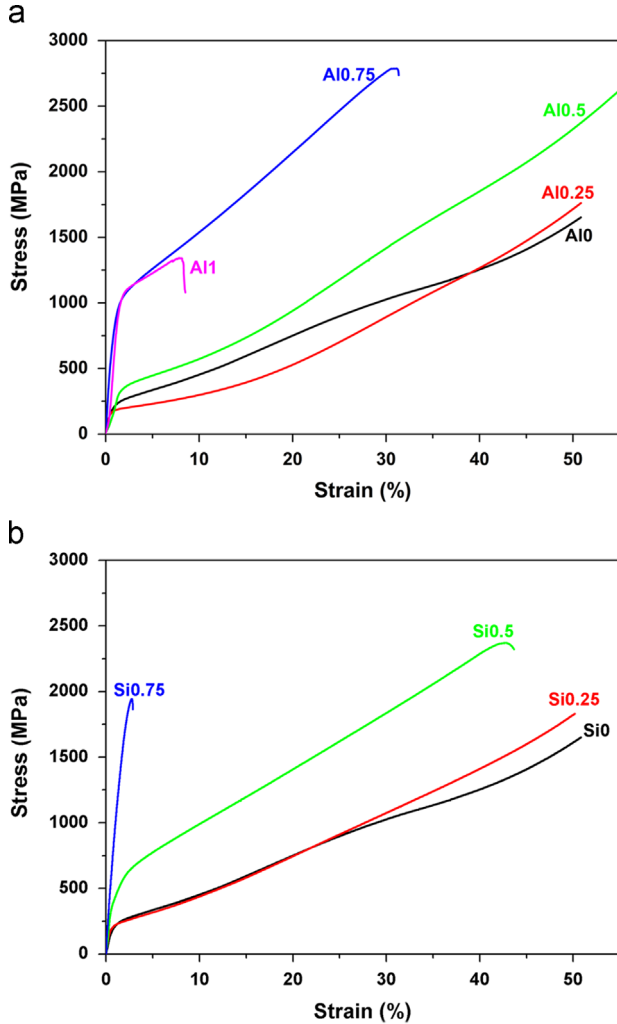


Fig. 5. Compressive engineering stress-strain curves of (a) Al_xCoFeNi and (b) CoFeNiSi_x alloys.

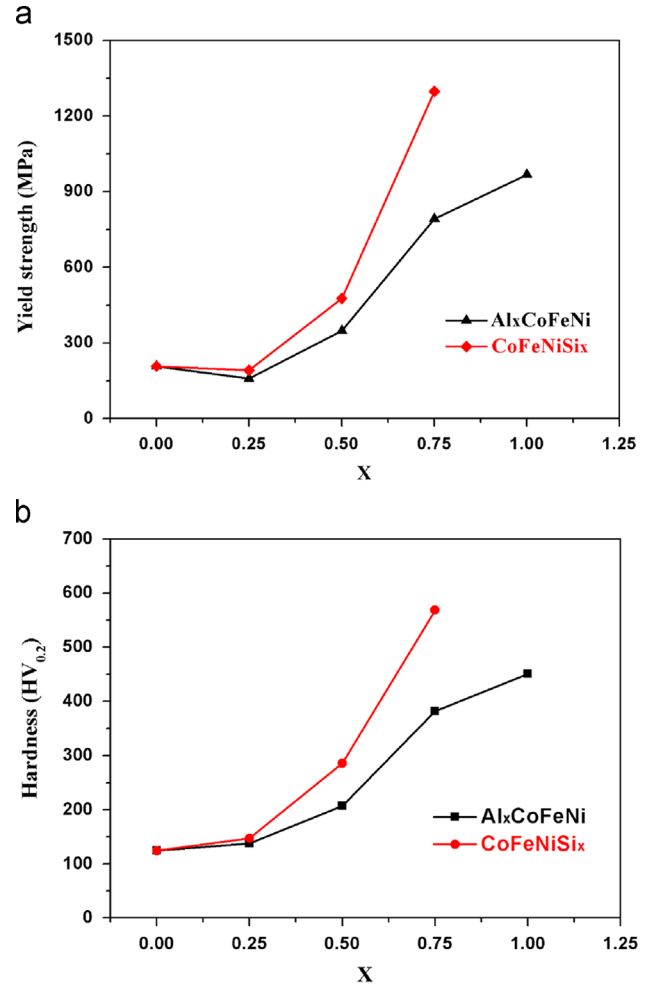


Fig. 7. Yield strength and hardness variation with different Al/Si content.

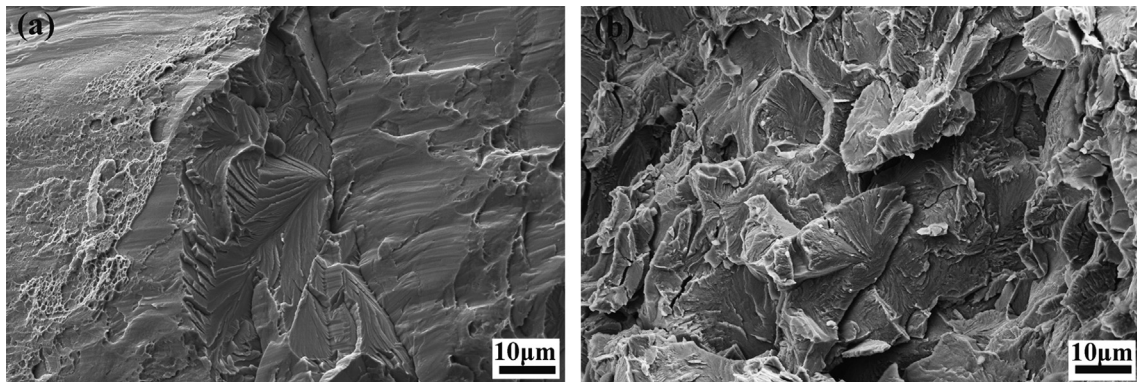


Fig. 6. SEM images of the fracture surface of (a) $\text{Al}_{0.75}\text{CoFeNi}$ and (b) $\text{CoFeNiSi}_{0.75}$ alloys.

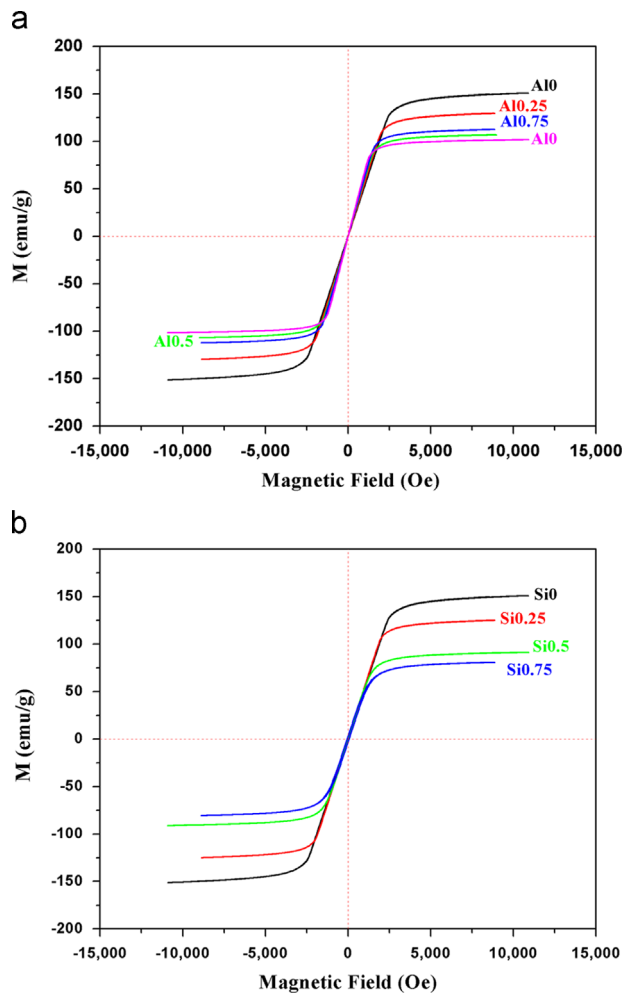


Fig. 8. Hysteresis loops of (a) Al_xCoFeNi and (b) CoFeNiSi_x alloys.

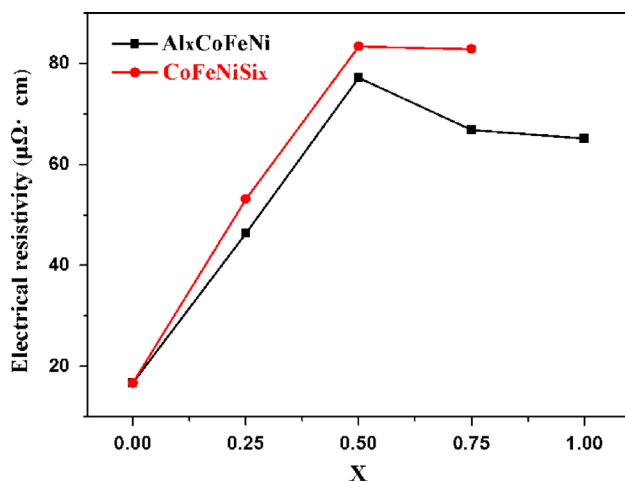


Fig. 9. The variation of electrical resistivity of Al_xCoFeNi and CoFeNiSi_x alloys.

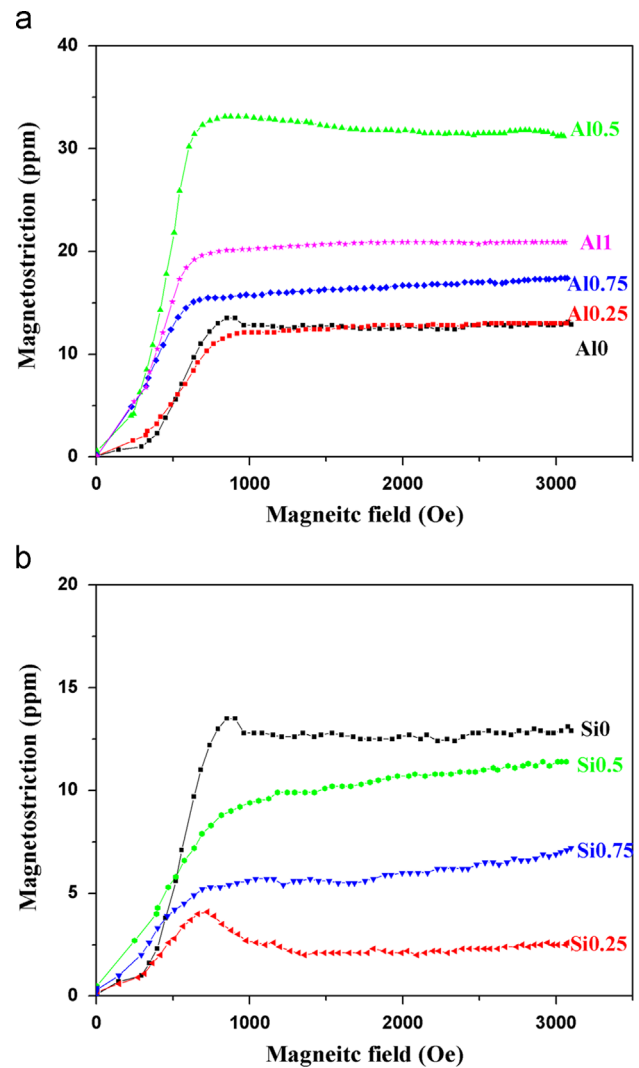


Fig. 10. The magnetostrictive curves of (a) Al_xCoFeNi and (b) CoFeNiSi_x alloys.

these elements, the magnetostriction coefficient can be made to be close to zero. It has been reported that Si addition can effectively reduce the magnetostriction coefficient of Fe–Ga alloy [26]. As shown in Fig. 10(b), similar beneficial effect can be found in CoFeNiSi_x alloys in this work, and the magnetostriction coefficient decreases significantly with a small amount of Si addition ($x=0.25$). In comparison with the previous studies, the content of Al and Si needs to be adjusted through fine composition design to obtain a soft magnetic HEA with balanced properties. It should be noted that the coercivity is not only influenced by the magnetostriction coefficient, it is also a microstructure-related parameter which could be influenced largely by grain size, the second phase, defects and processing technology, and so on [27,28]. As highlighted in the previous study, the ternary CoFeNi alloy in equal atomic percent which has the smallest coercivity (1.52 Oe) presents polycrystalline microstructure with the elements evenly distributed [16]. For Al_xCoFeNi alloys as shown in Fig. 11, H_c increases to the maximum value (4.31 Oe) when $x=0.5$, then decreases to 2.82 Oe at $x=1$. The variation trend of H_c with the Al content is similar to that of magnetostriction coefficient except for the case of Al1 alloy. Since the elements in Al1 alloy distribute relatively uniformly compared to Al0.5 and Al0.75 alloy (as both has a two phase structure), its resistance to magnetic domain wall movements becomes lower. But for CoFeNiSi_x alloys, the relation between H_c and λ_s is less clear. As shown by the data of H_c vs. Si content (Fig. 11), the value of H_c grows progressively when $x \leq 0.5$, then

parameters including magnetostriction coefficient and magnetic anisotropy coefficient have a decisive influence on the coercivity. Dimensional changes caused by magnetostriction coupled with stress would lead to magneto-elastic energy that could hinder magnetic domain from moving. In order to lower the coercivity, smaller magnetostriction coefficient should be obtained by adjusting the components [24]. For polycrystal metal, Fe shows a positive magnetostriction, while Ni is the opposite [25]. By tailoring the content of

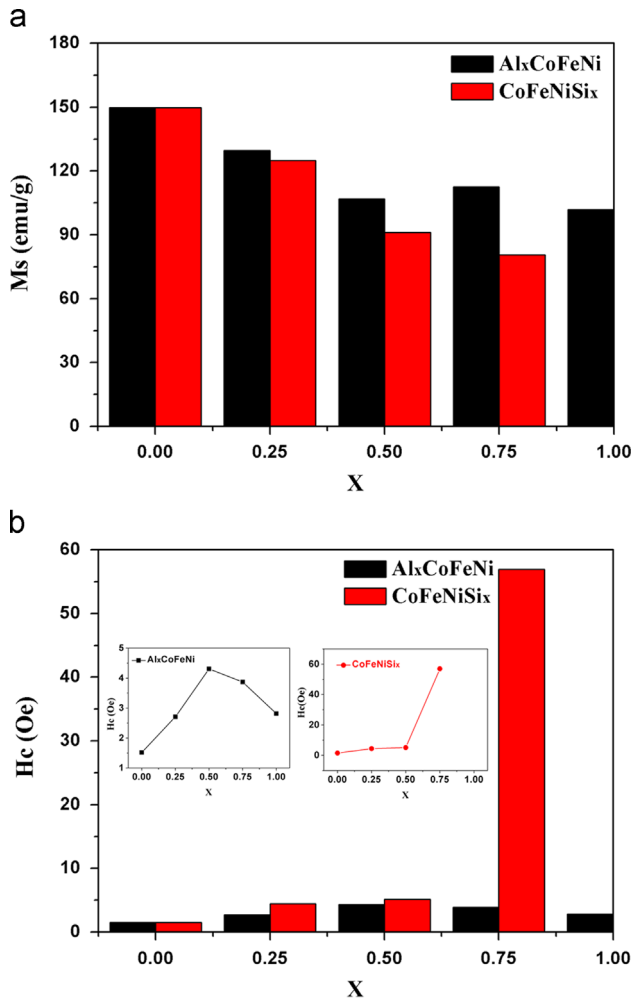


Fig. 11. The variation of M_s and H_c of $Al_xCoFeNi$ and $CoFeNiSi_x$ alloys.

increase sharply to 56.95 Oe at $x=0.75$. This is probably due to the fact that more precipitation in flocculent and needle-like morphology has formed in $Si_{0.75}$ alloy causing significant increase in H_c . Since H_c is very sensitive to microstructure, it is possible that, the coercivity can be tuned by suitable heat treatment, this can be a beneficial finding for the development of materials with optimum coercivity, which is to be investigated in future works.

4. Conclusion

In this study, the structures and properties of $Al_xCoFeNi$ and $CoFeNiSi_x$ alloys have been systematically investigated. It was found that the structure of $Al_xCoFeNi$ changes from single FCC to single BCC when x varies from 0 to 1 and exhibit typical dendritic microstructure. For the $CoFeNiSi_x$ alloys with the increase of Si addition, there are two types of Ni_3Si phases precipitate from the FCC matrix by spinodal decomposition. Mechanical tests showed that both Al and Si addition can increase the yield strength and reduce the ductility with the addition of Si showing a stronger influence. Typical ferromagnetic behavior is observed in these two alloy systems. The saturation magnetization of the alloys is found to decrease from 151.3 emu/g to 101.8 emu/g for $Al_xCoFeNi$ alloys; and from 151.3 emu/g to 80.5 emu/g for $CoFeNiSi_x$ alloys. Electrical property studies showed that the electrical resistivity increases with increasing Al or Si addition, in the case of $CoFeNiSi_x$, electrical resistivity increases from 16.7 $\mu\Omega$ cm to 82.89 $\mu\Omega$ cm. In addition,

both alloy systems undergo very small magnetostriction with the value lower than 35 ppm, and the data suggests that it is possible to achieve a lower magnetostriction by adjusting a small amount of Si content. The work implies that, in order to get the balance between mechanical and magnetic properties, systematic optimization of elements ratio of Al or Si is needed to achieve an optimum combination of properties.

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